

Assignment (Unit-2)

SECTION A (5 questions * 2 marks = 10 Marks)

1. Define encryption and explain the difference between symmetric and asymmetric encryption algorithms.
2. What is social engineering? Provide two examples of social engineering attacks and explain how they exploit human psychology.
3. Explain the term "phishing" and discuss two common indicators of a phishing email.
4. Define malware and provide examples of two types of malware. Briefly describe how each type of malware operates.
5. What is the purpose of a firewall in network security? Describe two different types of firewalls commonly used in cybersecurity.

SECTION B (3 questions * 7 marks = 21 Marks)

1. Discuss the importance of access control in maintaining cybersecurity. Explain the difference between discretionary access control and mandatory access control, providing examples of each.
2. Explain the concept of encryption key management. Discuss two key management practices that organizations can implement to enhance the security of their cryptographic keys.
3. What is a Denial of Service (DoS) attack? Describe three different types of DoS attacks and explain how they can be mitigated.

SECTION C (3 questions * 11 marks = 33 Marks)

1. Discuss the significance of threat intelligence in cybersecurity. Explain how organizations can leverage threat intelligence to enhance their security posture, providing examples of threat intelligence sources.
2. Explain the role of Intrusion Detection Systems (IDS) in network security. Compare and contrast host-based IDS and network-based IDS, discussing their advantages and limitations.
3. Outline the steps involved in developing a comprehensive cybersecurity policy for an organization. Discuss the key components that should be included in such a policy to effectively mitigate cybersecurity risks.